

# HTTP Methods Cheat Sheet

Prepared by: Naveen Automation Labs

LinkedIn: <https://www.linkedin.com/in/naveenkhunteta/>

YouTube: <https://www.youtube.com/@naveenautomationlabs/>

## Overview of HTTP Methods

| Method  | Purpose                                     | Cacheable?           | Request Content | Safe | Idempotent |
|---------|---------------------------------------------|----------------------|-----------------|------|------------|
| GET     | Retrieve a representation of a resource     | Yes                  | Not recommended | Yes  | Yes        |
| HEAD    | Same as GET but without response body       | Yes                  | Not recommended | Yes  | Yes        |
| POST    | Process the enclosed representation         | Only with conditions | Yes             | No   | No         |
| PUT     | Create or replace the state of a resource   | No                   | Yes             | No   | Yes        |
| DELETE  | Remove the resource's association           | No                   | Not recommended | No   | Yes        |
| CONNECT | Establish a tunnel to the server            | No                   | No              | No   | No         |
| OPTIONS | Get information about communication options | No                   | Optional        | Yes  | Yes        |
| TRACE   | Perform a message loop-back test            | No                   | No              | Yes  | Yes        |

## Detailed Method Descriptions

### GET

- **Primary purpose:** Information retrieval
- **Response codes:** Any status code can be returned
- **Typical success response:** 200 OK
- **Can be modified by:** Range header field (for partial content)
- **Cacheable:** Yes
- **Special features:**
  - Focus of most performance optimizations
  - Can request partial content with Range header
- **Best practices:**
  - Don't include sensitive data in URI

- Don't send request content to general servers

## HEAD

- **Primary purpose:** Obtain metadata without transferring representation data
- **Response codes:** Same as GET
- **Response headers:** Should be identical to a GET response
- **Cacheable:** Yes
- **Special features:**
  - Can affect cached responses to GET
  - Useful for testing links or finding recent modifications
- **Best practices:**
  - Don't send request content to general servers

## POST

- **Primary purpose:** Process enclosed representation according to resource's semantics
- **Common uses:**
  - Form submission
  - Posting to message boards/blogs
  - Creating new resources
  - Appending data to existing resources
- **Response codes:** Almost any status code
- **Typical success response:**
  - 200 OK (processed)
  - 201 Created (new resource created)
  - 303 See Other (redirect to new resource)
- **Cacheable:** Only with explicit freshness info and matching Content-Location
- **Best practices:**
  - Use for non-idempotent operations
  - Use when operation may have side effects

## PUT

- **Primary purpose:** Create or replace resource state with enclosed representation
- **Response codes:**
  - 201 Created (new resource created)
  - 200 OK or 204 No Content (resource updated)

- 409 Conflict or 415 Unsupported Media Type (incompatible representation)
- **Cacheable:** No
- **Special features:**
  - Idempotent (multiple identical requests have same effect as single request)
  - Can support partial updates with Content-Range
- **Best practices:**
  - Use when client knows resource URI
  - Don't use for operations with variable outcomes

## DELETE

- **Primary purpose:** Remove association between resource and its functionality
- **Response codes:**
  - 202 Accepted (will likely succeed but not yet enacted)
  - 204 No Content (enacted with no further info)
  - 200 OK (enacted with status representation)
- **Cacheable:** No
- **Best practices:**
  - Limited availability (primarily for authoring environments)
  - Don't send request content to general servers

## CONNECT

- **Primary purpose:** Establish a tunnel to destination server
- **Request target format:** host  
(unique to this method)
- **Response codes:**
  - 2xx (tunnel established, switching to tunnel mode)
  - Any other (tunnel not formed)
- **Cacheable:** No
- **Special features:**
  - Often used with TLS for secure connections
  - Tunnel closes when either side closes connection
- **Best practices:**
  - Primarily for proxy requests
  - Should be restricted to known/safe ports

## OPTIONS

- **Primary purpose:** Get information about communication options
- **Request target:**
  - "\*" (applies to server in general)
  - Specific URI (applies to that resource)
- **Response headers:** Should indicate optional features (e.g., Allow)
- **Cacheable:** No
- **Best practices:**
  - Use to test server capabilities
  - May include content with valid Content-Type

## TRACE

- **Primary purpose:** Application-level loop-back for testing/diagnostics
- **Response:** Mirror of the request message
- **Cacheable:** No
- **Special features:**
  - Can use Max-Forwards to limit request chain length
  - Via header provides trace of request chain
- **Best practices:**
  - Never include sensitive data
  - Never send content in request

## Key Differences Table

| Feature                | GET            | HEAD            | POST          | PUT                      | DELETE           |
|------------------------|----------------|-----------------|---------------|--------------------------|------------------|
| Resource retrieval     | Yes            | Metadata only   | No            | No                       | No               |
| Resource modification  | No             | No              | Possible      | Yes                      | Yes              |
| Request body expected  | No             | No              | Yes           | Yes                      | No               |
| Response body returned | Yes            | No              | Yes           | Yes                      | Optional         |
| Cacheable              | Yes            | Yes             | Conditionally | No                       | No               |
| Target specified as    | Resource URI   | Resource URI    | Resource URI  | Resource URI             | Resource URI     |
| Idempotent             | Yes            | Yes             | No            | Yes                      | Yes              |
| Safe (read-only)       | Yes            | Yes             | No            | No                       | No               |
| Primary use case       | Data retrieval | Header checking | Multi-purpose | Resource creation/update | Resource removal |

| Feature                | CONNECT       | OPTIONS              | TRACE          |
|------------------------|---------------|----------------------|----------------|
| Resource retrieval     | No            | No                   | No             |
| Resource modification  | No            | No                   | No             |
| Request body expected  | No            | Optional             | No             |
| Response body returned | Data tunneled | Optional             | Request mirror |
| Cacheable              | No            | No                   | No             |
| Target specified as    | host          | Resource URI or ""   | Resource URI   |
| Idempotent             | No            | Yes                  | Yes            |
| Safe (read-only)       | No            | Yes                  | Yes            |
| Primary use case       | Tunneling     | Capability discovery | Diagnostics    |

Common Response Status Codes by Method

| Method  | Common Success Status Codes          | Common Error Status Codes                                                |
|---------|--------------------------------------|--------------------------------------------------------------------------|
| GET     | 200 OK, 206 Partial Content          | 404 Not Found, 403 Forbidden                                             |
| HEAD    | 200 OK                               | 404 Not Found, 403 Forbidden                                             |
| POST    | 200 OK, 201 Created, 303 See Other   | 400 Bad Request, 404 Not Found, 422 Unprocessable Entity                 |
| PUT     | 200 OK, 201 Created, 204 No Content  | 400 Bad Request, 404 Not Found, 409 Conflict, 415 Unsupported Media Type |
| DELETE  | 200 OK, 202 Accepted, 204 No Content | 404 Not Found, 403 Forbidden                                             |
| CONNECT | 200 OK                               | 400 Bad Request, 403 Forbidden, 405 Method Not Allowed                   |
| OPTIONS | 200 OK                               | 403 Forbidden, 404 Not Found                                             |
| TRACE   | 200 OK                               | 403 Forbidden, 405 Method Not Allowed                                    |

Prepared by: Naveen Automation Labs

LinkedIn: <https://www.linkedin.com/in/naveenkhunteta/>

YouTube: <https://www.youtube.com/@naveenautomationlabs/>